TuteDude

Capstone Project- 1

EdTech Platform User Engagement and Learning Behavior Analysis

Objective

EdTech platforms rely on data analytics to understand user engagement, learning behavior, and course completion patterns. Business analysts use these insights to improve learning outcomes, optimize platform features, and increase user satisfaction.

The objective of this capstone project is to perform a comprehensive analysis of an online learning platform dataset and generate meaningful insights about user behavior and platform performance.

Business Scenario

You are working as a **Business Analyst for an EdTech SaaS platform** that offers online learning programs to users across different demographics.

The management team wants to better understand how users interact with the platform and what factors influence learning engagement, course completion, and user satisfaction.

Your task is to analyze the dataset and generate insights that can help the company improve the overall learning experience and platform effectiveness.

Dataset to be used:-

online learning platform usage dataset

Instructions

Step 1: Dataset Understanding

Explore the dataset and understand its structure.

Identify:

- number of rows and columns
- data types of each variable
- key attributes describing user behavior

Document your understanding of the dataset.

Step 2: Data Preparation

Prepare the dataset for analysis.

Tasks may include:

- checking for missing values
- verifying data consistency
- converting columns to appropriate data types

Ensure the dataset is clean and ready for analysis.

Step 3: Exploratory Data Analysis

Analyze user behavior patterns using descriptive statistics.

Examples of analysis may include:

- distribution of users across age groups
- hours spent on learning per week
- platform usage patterns
- preferred learning content types

Summarize key observations from the analysis.

Step 4: Engagement Analysis

Analyze how actively users interact with the platform.

Possible analyses include:

- average learning hours per user group
- relationship between learning hours and completion rate
- comparison of engagement across different platforms or devices

Identify which groups appear most engaged on the platform.

Step 5: Course Completion Analysis

Evaluate factors that influence course completion.

Examples:

- which user segments show higher completion rates
- relationship between engagement and completion rate
- impact of learning purpose on completion behavior

Highlight patterns that explain course success or drop-off.

Step 6: User Satisfaction Analysis

Analyze satisfaction scores to understand user experience.

Possible insights include:

- relationship between completion rate and satisfaction score
- differences in satisfaction across user groups
- patterns in satisfaction based on platform or device usage

Step 7: Data Visualization

Create visualizations that effectively communicate insights.

Examples may include:

- distribution charts for engagement levels
- comparison charts for completion rates
- satisfaction score visualizations
- platform usage patterns

Visualizations should clearly explain the story behind the data.

Step 8: Key Business Insights

Summarize **5–7 key insights** discovered during the analysis.

Examples may include:

- which users engage the most with the platform
- which factors influence completion rates
- patterns that affect user satisfaction

Step 9: Business Recommendations

Provide **3–5 recommendations** that could help improve the platform.

Examples may include:

- improving engagement strategies
- enhancing course completion support
- optimizing platform features for better user experience

Submission Guidelines

Your submission must include:

- Screenshots showing analysis steps
- Visualizations created during the analysis

- A document explaining insights and recommendations
- Python notebook file (.ipynb)

Output File Requirements in Task Submission

Your submission folder may contain the following files:

1. Screenshots
2. Image
3. Doc file
4. PDF file
5. Code file / md file
6. Excel
7. Python notebook file (.ipynb)

Final Submission Process (IMPORTANT)

1. Collect all required output files.
2. Place all files inside **one folder**.
3. Name the folder clearly according to the task.
4. Compress the **entire folder into a ZIP file**.
5. Submit the **ZIP folder**.

⚠️ Do not submit individual files. Only ZIP folders will be accepted.

Watch this to learn how to create ZIP file
https://youtu.be/c7x-Q3AQEgw?si=_mS4c7OXADQ7DroE

If you have any doubt, you can directly contact your mentor.